

Problem C. Permutation Inversions

Input file: standard input
Output file: standard output
Time limit: 3 seconds
Memory limit: 1024 megabytes

There is an unknown permutation $p = (p_1, p_2, \dots, p_n)$ of the integers $1, 2, \dots, n$. You are given m pieces of information about this permutation.

Each piece of information consists of an interval $[l_i, r_i]$ and a permutation $q_{i,1}, q_{i,2}, \dots, q_{i,r_i-l_i+1}$ of the indices in this interval. It states that

$$p_{q_{i,1}} < p_{q_{i,2}} < \dots < p_{q_{i,r_i-l_i+1}}.$$

Your task is to find a permutation that satisfies all the given information and has the minimum possible number of inversions. If no such permutation exists, output -1 .

If several optimal permutations exist, output any of them.

The number of inversions of a permutation p is the number of pairs of indices (x, y) such that

$$1 \leq x < y \leq n \quad \text{and} \quad p_x > p_y.$$

Input

The first line contains a single integer T ($1 \leq T \leq 10^6$) — the number of test cases.

For each test case:

- The first line contains two integers n ($1 \leq n \leq 10^6$) and m ($1 \leq m \leq 10^6$) — the size of the permutation and the number of constraints.
- Each of the next m lines describes a constraint. The i -th constraint starts with two integers l_i and r_i ($1 \leq l_i \leq r_i \leq n$), followed by $r_i - l_i + 1$ integers $q_{i,1}, q_{i,2}, \dots, q_{i,r_i-l_i+1}$, which form a permutation of the integers from l_i to r_i .

It is guaranteed that $1 \leq \sum n \leq 10^6$, $1 \leq \sum m \leq 10^6$, and $1 \leq \sum_{i=1}^m (r_i - l_i + 1) \leq 10^6$ over all test cases.

Output

For each test case, print a single line:

- If no satisfying permutation exists, print -1 .
- Otherwise, print n space-separated integers $p_1, p_2, \dots, p_n$ representing a valid permutation that minimizes the number of inversions.

If there are multiple optimal permutations, print any of them.

Example

standard input	standard output
4	1 3 2 4
4 1	-1
2 4 3 2 4	1 3 4 2 5
3 2	2 1 4 3
1 3 1 2 3	
1 3 3 2 1	
5 2	
2 4 4 2 3	
2 4 4 2 3	
4 2	
2 2 2	
1 4 2 1 4 3	